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Assessing the Learning Needs of Maternal and Child Health Professionals to Teach Health Promotion

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Objective: To assess skills and preferred learning formats in teaching health promotion (HP). *Methods*: A self-administered needs assessment of Maternal and Child Health (MCH) educators from multiple disciplines was conducted on a convenience sample taken nationally via e-mail using Dillman's method. Respondents rated ability to use, and desire to improve skills in, different teaching strategies (brainstorming, case method, collaborative learning, mini-presentation, reflective exercise, role play) and health concepts (partnership, communication, HP/illness prevention, time management, education, advocacy). Preferred learning formats were assessed with 5-point Likert scale and were analyzed using ANOVA. *Results*: Fifty-seven percent of respondents ($n = 180$) taught in an urban setting, 26% suburban, and 17% rural. Most taught at academic health centers (35%), public health clinics (25%), or hospitals (17%). Seventy-five percent were female; average age was 42 years (SD-9.1 years). Specific disciplines showed no major difference in mean responses compared with others. The greatest barriers to integrating HP into teaching were time (82%) and budget (58%). Although a majority of all respondents felt comfortable in their abilities to use the teaching strategies and concepts, an equal percentage still wanted to improve these skills. One-third of respondents had experience using web-based study; 64% of them indicated web-based study as their preferred method of continuing education. *Conclusions*: While a majority of MCH educators felt confident using various teaching strategies to teach the integration of HP into practice, most still wished to improve their personal skills. Use of an inexpensive, time-efficient modality to access and learn to teach HP was appealing to respondents across disciplines.

KEY WORDS: medical education; distance learning; health promotion; faculty development; continuing education.

INTRODUCTION

Chronic conditions and behavioral risks, such as violence, smoking, drugs, alcohol, sexual intercourse, and poor eating habits, threaten the health and well-being of America's children (1–7). These behaviors may be preventable if addressed through meaningful health promotion (HP) efforts. The increasing prevalence of these health concerns warrants a shift in the way in which maternal and child health (MCH) professionals, including doctors, nurses, dentists, social workers, and public health professionals, are trained to deliver health promotion content and in the steps taken to ensure their patients' health. Changes in the nature of health care delivery and funding for teaching institutions may influence the ways in which MCH professional training programs are implemented; focusing on clinical service may limit both curriculum and faculty development in this area. More formal, effective, and standardized clinician–teacher training programs could help to fill this gap.

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The mission of the *Bright Futures Guidelines for Health Supervision of Infants, Children, and Adolescents* is "to promote and improve the health and well-being of infants, children, adolescents, families, and communities" (8) through the development of educational materials for health professionals and families (www.brightfutures.org or brightfutures.aap.org). It can help child health professionals to focus on a child in the context of his/her family and community as they incorporate health promotion into their clinical practices in a time-efficient manner. A *Bright Futures* Health Promotion Workgroup, composed of pediatricians, educators, parents, and nurses, was formed in 1998 with the charge of developing a HP curriculum that addresses the key skills, knowledge, and attitudes necessary for child health professionals to integrate *Bright Futures'* goals into daily practice.

Pediatrics in Practice, A Health Promotion Curriculum for Child Health Professionals presents six core concepts of health promotion (partnership, communication, HP/illness prevention, time management, education, and advocacy) and six teaching strategies (brainstorming, case method, collaborative learning, mini-presentation, reflective exercise, and role play) that are available in both published textbook and distance-education formats (e.g., printable PDFs for tangible references when users are away from the computer). This allows for broader access and dissemination, ease of navigation, excellent search capabilities, and a clear presentation of the curriculum.

One key advantage to using Internet technology for distance learning (DL) is the ability to overcome barriers of space and time in teaching and learning (9). Several other advantages of DL include broader dissemination to a global audience, shared resources, less reliance on personnel/teachers, the possibility of adaptation to learning styles, easy updating of content and archival capabilities, lower development and operating costs compared to broadcast, support for self-directed learning, and speed and efficiency over other learning modes (9). DL is also an effective learning tool for both medical professionals (10–17) and patients (18–22).

To date, there have been very few studies evaluating distance learning for educators. Langlois and Thach (23) discussed the difficulty in providing faculty development to community-based faculty who are dispersed over wide geographic areas. Steinert *et al.* (24) reported failure in attempting to use the Internet for an on-line discussion group to promote faculty development. They cite as possible barriers time, competing demands, and the perception of the discussion group as unnecessary.

The purpose of this study is to determine self-identified MCH educators' needs in delivering HP content in their professional settings, to identify their skills in using various teaching strategies to do so, and to assess their personal learning preferences. We hypothesized that designing and delivering effective, efficient faculty development programs, which improve faculty skills in teaching health promotion, would be valuable to MCH educators' professional development. MCH educators were defined as those professionals involved in any aspect of MCH education.

METHODS

A self-administered survey instrument was developed to assess the training needs of MCH educators in geographically diverse locations. A link to the survey was distributed via e-mail, and embedded in an attached document, to one MCH educator of every available discipline at the 11 *Bright Futures* Health Promotion Workgroup members' sites around the country. Members of our Health Promotion Workgroup identified contact people as educators in medicine, nursing, dentistry, social work, and public health. These secondary contacts were asked to complete the survey and "forward the link via e-mail to all members of their faculty involved in any aspect of MCH education." Dillman's method (25) was used in the survey administration to maximize survey response rates. This method is used to achieve high response rates for self-administered surveys by using a series of four contacts by sending: 1) a prenotice letter, 2) a questionnaire, 3) a reminder, 4) resending the questionnaire.

Survey Development

This survey was designed to assess the curricular and educational needs of MCH educators. The individual items were developed from a review of the literature (26–28), an examination of existing instruments (29–36), and informal consultation with state MCH educators, directors of child health groups/organizations, and professionals/residents who care for children. The survey was created after examination of several instruments including the American Academy of Pediatrics Continuing Medical Education needs assessment in 2000

(33), Office of Educational Development of the University of North Carolina School of Medicine needs assessment (34, 35) and the Maternal and Child Health Bureau pediatric residency survey (36).

The instrument contained questions about the respondents' professional background(s) and experiences as educators. Experiences and preferences concerning continuing education were assessed. Likert-scale items gauged confidence in the respondents' ability to use each of the teaching strategies: brainstorming, case method, collaborative learning, mini-presentation, reflective exercise, role play, as well as desire for improvement in each of these areas. Additionally, respondents rated their confidence and desire for improvement on 5-point Likert scale in six health promotion core concepts: partnership, communication, HP/illness prevention, time management, education, and advocacy. Items in the instrument also measured the most common computer-based technologies used by MCH educators and their baseline computer skills. The instrument was pilot tested with MCH professionals at the primary site, Children's Hospital Boston.

Analyses

Summary statistics (mean and standard deviation) were calculated for all variables of interest. Chi-square tests were used to compare males and females, and *t* tests were used to compare ages. Analysis of Variance (ANOVA) tests were used to compare across the following seven mutually exclusive disciplines: dentistry only, medicine only, nursing only, public health only, social work only, public health/other, and miscellaneous. When the ANOVA test was statistically significant, post hoc Tukey Honestly Significant Difference tests were conducted to determine if disciplines differed significantly. Statistical significance was defined as $p < 0.001$ using the Bonferroni correction for multiplicity.

RESULTS

Demographics

Seventy-five percent of all survey respondents ($n = 180$) were female; the average age of respondents was 42 years (SD-9.1 years). A majority had backgrounds in nursing (41%), public health (27%), or medicine (27%). The remaining disciplines were dentistry (14%), social work (8%), or "other" (6%) backgrounds. Fifty-seven percent of respondents taught in an urban location, 26% suburban, and 17% rural. Ten percent of respondents had degrees in education. Survey respondents were responsible for teaching a wide variety of disciplines, including medicine (33%), public health (20%), registered nurses (18%), dentists (14%), nurse practitioners (12%), physician assistants (6%), social workers (5%) and others (22%) such as childcare, CPR, injury prevention, nutrition, and early intervention. Most survey respondents taught in an academic health center (35%), a public health clinic (25%), or a hospital (17%). Others taught in a community health center (2%), private practice (2%), or other locations (18%).

Needs and Abilities of MCH Educators

A variety of ways to participate in continuing education programs were reported. Many respondents reported having attended teaching seminars or lectures (46%) or teaching workshops (41%) for continuing education. Although day-long seminars or lectures were the preferred methods (Fig. 1), over one-third preferred web-based study for continuing education. A majority of respondents agreed or strongly agreed that they were confident in their ability to use the six teaching strategies, yet they still would like to improve their skills in these same areas (Fig. 2). The majority of respondents agreed or strongly agreed that they were confident in their ability to teach the six health promotion concepts, and they would also like to improve their skills in the same concepts (Fig. 3). In all questions, 6% or less of respondents indicated that they would *not* like to improve their ability to teach each of these concepts ("disagree" or "strongly disagree"). Eighty-two percent of respondents indicated that time constraints were a barrier to integrating health promotion information into teaching settings, 58% indicated limited budget, and 30% indicated lack of resources.

The majority of survey respondents (62%) considered themselves knowledgeable about the Internet, with 28% considering themselves beginners, 4% novice, and 6% expert. Respondents had a wide range of Internet connections, while 39% did not know what type of connection they had. Ninety-six percent of respondents indicated "strongly agree" or "agree" to their confidence in using e-mail, 88% word processing, 80% search engines, 78% "surfing" the Internet, 75% CD-ROM,

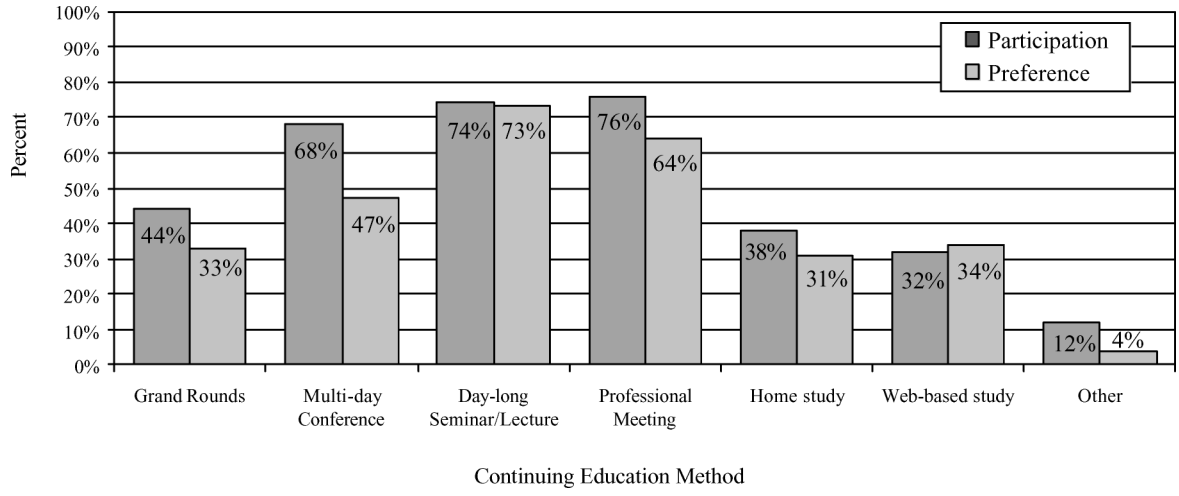


Fig. 1. Respondents' participation and preferred methods in continuing education programs ($n = 180$).

71% uploading/downloading, 62% PowerPoint, 21% chat rooms, and 15% video streaming.

Comparisons According to Discipline, Gender, and Age

There were no significant differences according to gender or age. Comparing results across the seven mutually exclusive disciplines, those who classified themselves as social workers only were found to have less confidence in HP/illness prevention than those in the medicine only discipline ($p < 0.001$), nursing only ($p < 0.001$), and public health/other ($p < 0.001$). Other results were not statistically significant across disciplines.

DISCUSSION

There was strong interest among MCH professionals in our study for improving skills in various teaching strategies. Equipped with this knowledge, professionals should have the necessary tools

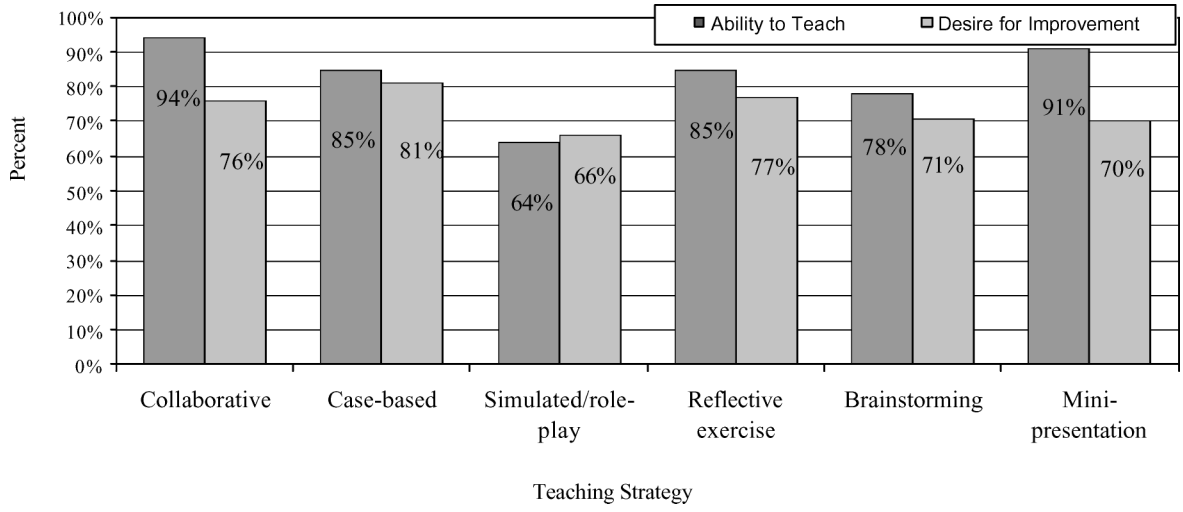


Fig. 2. Percentage of respondents indicating "strongly agree" or "agree" in their ability to teach health promotion strategies and their desire for improvement teaching these strategies ($n = 165$–172).

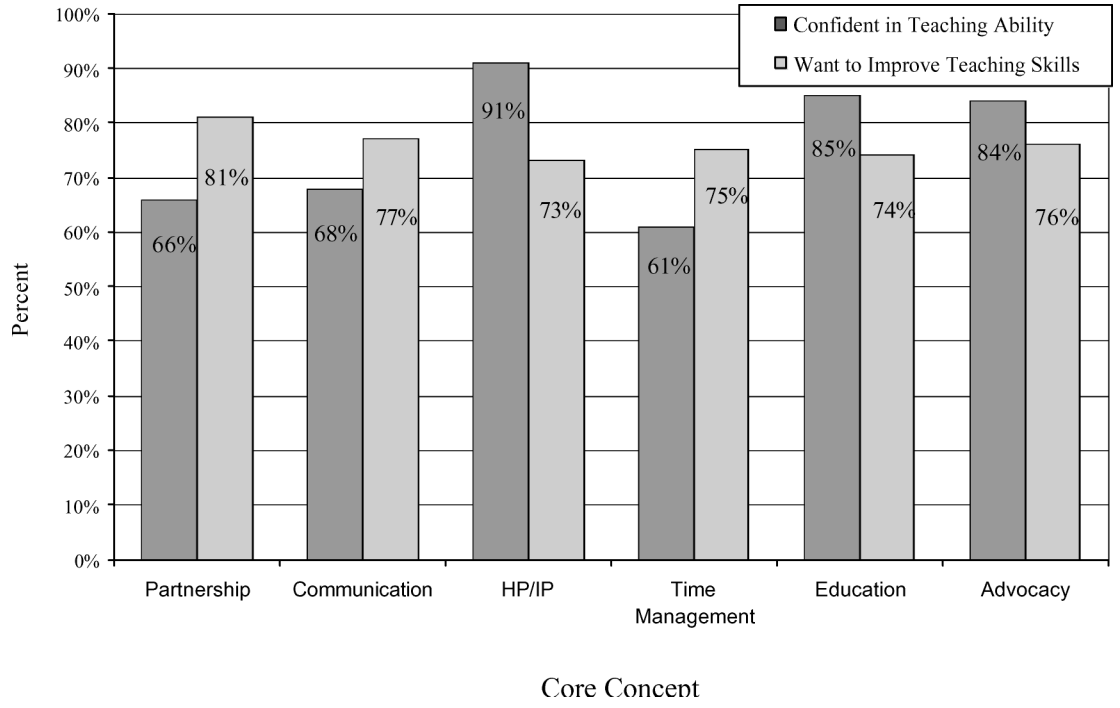


Fig. 3. Percentage of respondents indicating "strongly agree" or "agree" in their ability to use teaching strategies and their desire to improve their skills in the strategies ($n = 165–172$).

to further disseminate HP content. MCH professionals also expressed a strong interest in improving skills in several HP concepts. While respondents with a background in social work reported less confidence than others, all had an equally strong desire to improve their skills in the teaching strategies and core concepts. Further research examining disciplinary responses toward teaching health promotion may reveal why social workers indicated less confidence in teaching HP/illness prevention.

While increasingly used, our method of survey distribution has several limitations. We were unable to ascertain an accurate response rate because we do not know how many people received the survey but did not return it. Further, by distributing the survey to facilitators at primary sites who, in turn, distributed it to their various constituencies on site, we were not able to garner a random sample. Moreover, the use of an e-mail-based survey and the Internet may have overrepresented respondents with confidence and skills in Internet use. By asking people via e-mail about their abilities to use the Internet, we also excluded people who do not use either e-mail or the Internet. Therefore, we do not have data from health professionals who are not currently using the Internet but may be interested in a user-friendly distance-learning program. A 1999 study conducted in England (37) found postal surveys to more effectively garner a response rate than e-mail surveys with a world wide web form (72 vs. 19%). Also, the use of self-reporting may not accurately reflect how respondents truly behave. Because we surveyed a wide variety of MCH educators, we were unable to determine which subsets of educators are most interested in our curriculum. Despite these limitations, respondents showed a strong interest in distance learning and health promotion, consistent with prior studies (23, 24, 38).

Distance education is an innovative and timely approach to training learners in an effective and self-directed manner. The Education Scholar (39), one of few faculty development programs for health professions educators, is described as "a comprehensive web-based program for the advancement of teaching skills, professional scholarship, and instructional improvement for educators in the health professions." Research has not yet been done to evaluate this program, and very little research has been conducted on using distance education for faculty development overall. Many experts offer that online learning has advantages, but barriers such as inadequate equipment (40), perceived need (24), or the overwhelming

amount of information provided by the Internet (41) may detract from its use.

Several other distance education programs have further elucidated these barriers. A Preceptor Development Program (PDP) was established at the Mountain Area Health Education Center in Asheville, North Carolina in order to bring faculty development to community-based preceptors (23). Similar to the *Pediatrics in Practice* curriculum, the PDP has several topics and teaching strategies. While the web page had many visitors, no preceptors applied for CME credit. The program developers hypothesized that this result may have been due to limitations of the software: users were required to download a free software program before they were able to view modules.

Steinert *et al.* (24) implemented a smaller scale online workshop for medical faculty where an e-mail listserv was created after a one-day workshop on small-group teaching. Most participants liked the idea of an e-mail listserv follow-up; but only two individuals (8%) responded to the listserv after several attempts. The authors concluded that time and other demands were barriers to distance learning utilization, and faculty did not see the need for discussing small-group teaching online. They also mentioned many of their faculty may not have been comfortable with online learning.

The Maine Medical Center Family Practice Residency Program designed a lengthy multifaceted intervention for faculty development, aiming to increase knowledge and skills in the use of information technology (38). The goal of the project was to create a community of learners by changing the culture of the department of family practice at the Maine Medical Center; it resulted in faculty members indicating they were more comfortable/knowledgeable in informatics skills.

Conducting this online needs assessment revealed a need for and interest in time effective health promotion content on the part of MCH educators from various disciplinary backgrounds across the United States. While future surveys might assess content-specific needs, the concept of distance education is promising. These findings have helped to inform the development of a *Bright Futures*-based health promotion distance learning program *Pediatrics in Practice* (www.pediatricsinpractice.org), available in paper-based and online formats. It presents MCH educators with a standardized faculty development program designed to improve their abilities to teach health promotion content in the spirit of the *Bright Futures* framework. In an effort to build a community of learners, special attention is directed toward the core support and reinforcement components, including the posting and exchange of resources between users. *Pediatrics in Practice* promotes self-directed learning through the utilization of diverse techniques including reflection, self-assessment, journaling, and interactive "progress checks" designed to build cognitive skills in health promotion.

CONCLUSION

Health promotion is an increasingly integral part of maternal and child health care and was reflected in MCH educators' interest in learning more about the core concepts of health promotion. Use of an inexpensive, time-efficient modality to access and learn to teach HP was appealing to respondents across disciplines. An online health promotion curriculum can support MCH educators in building the necessary skills, knowledge, and attitudes among their future learners to provide quality care for children and families.

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